

CATANIA FONTANAROSSA, Italy

WMO: 164600

Lat: 37.467N

Lon: 15.066E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.4	2.8	-3.2	2.9	7.7	-1.8	3.3	7.2	11.9	12.8	10.5	13.2	2.9	240	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	34.7	22.8	32.9	22.8	31.3	22.5	26.1	30.0	25.2	29.5	24.3	28.9	4.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.2	28.0	24.0	18.9	27.5	23.0	17.7	26.9	81.1	29.6	76.8	29.2	73.3	28.9	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.9	8.5	7.4		-0.9	39.7	1.0	2.8	-1.6	41.7	-2.1	43.3	-2.7	44.8	-3.4	46.8	
				-1.7	27.1	1.0	1.9	-2.4	28.5	-3.0	29.6	-3.6	30.7	-4.3	32.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	10.4	10.6	12.3	14.7	18.5	22.8	25.5	26.0	23.3	19.6	15.3	11.8
	DBStd	6.07	2.45	2.41	2.49	2.16	2.41	2.57	2.00	1.94	2.28	2.32	2.61	2.51
	HDD10.0	62	24	19	6	1	0	0	0	0	0	0	1	11
	HDD18.3	1102	247	217	188	109	27	1	0	0	1	14	96	203
	CDD10.0	2836	36	36	77	143	264	383	482	497	398	296	159	66
	CDD18.3	834	0	0	1	1	32	133	223	239	149	52	4	0
	CDH23.3	8081	0	1	12	14	234	1257	2533	2641	1146	229	14	1
	CDH26.7	3014	0	0	4	2	45	408	1077	1132	318	28	1	0
Wind	WSAvg	3.4	3.4	3.6	3.7	3.7	3.5	3.4	3.4	3.2	3.3	3.1	3.2	3.3
Precipitation	PrecAvg	564	75	54	49	36	19	5	6	9	40	127	62	86
	PrecMax	927	258	147	157	154	115	19	54	53	209	371	228	405
	PrecMin	221	11	3	3	1	0	0	0	0	1	9	3	1
	PrecStd	201	65	39	36	34	25	7	12	13	42	99	62	83
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	20.8	24.1	25.0	30.7	35.8	37.7	37.6	34.1	29.5	25.1	21.1
		MCWB	13.8	13.4	14.9	15.8	18.8	20.9	23.4	23.6	21.4	20.2	18.3	15.6
	2%	DB	18.0	18.8	20.9	22.9	27.2	32.1	34.2	34.5	31.1	27.1	23.0	19.1
		MCWB	12.4	12.6	13.8	15.1	18.5	21.5	23.0	23.3	22.1	19.9	17.5	14.1
	5%	DB	16.7	17.1	19.0	21.1	25.8	30.2	32.7	32.9	29.8	25.8	21.8	17.9
		MCWB	12.2	12.1	13.1	14.8	18.1	21.3	22.7	23.3	21.6	19.5	17.1	13.7
	10%	DB	15.2	15.9	17.8	20.0	24.1	28.9	31.1	31.2	28.2	24.3	20.2	16.8
		MCWB	11.5	11.7	12.7	14.5	17.4	20.7	22.3	23.1	21.3	19.2	16.1	13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	15.1	16.1	17.9	21.2	25.4	26.9	27.8	26.0	23.5	20.8	17.0
		MCDB	17.3	17.7	20.4	20.8	25.2	29.7	31.5	30.7	28.7	25.2	22.5	19.4
	2%	WB	14.1	14.0	15.0	16.9	20.1	23.9	25.6	26.4	24.6	22.1	19.2	15.7
		MCDB	16.6	16.7	18.3	20.3	24.5	28.4	30.0	30.1	27.7	24.7	21.6	17.9
	5%	WB	13.1	13.2	14.2	16.0	19.3	22.9	24.7	25.6	23.6	21.1	18.0	14.8
		MCDB	15.8	15.9	17.4	19.2	23.7	27.9	29.4	29.8	27.1	24.0	20.4	17.1
	10%	WB	12.1	12.3	13.5	15.4	18.4	21.9	23.9	24.7	22.7	20.2	16.9	13.6
		MCDB	14.8	15.4	16.7	18.5	22.8	27.2	29.0	29.5	26.4	23.4	19.5	16.1
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.5	10.7	10.6	11.1	11.4	11.7	11.2	10.3	9.7	9.8	9.6
		MCDBR	11.7	12.1	13.0	12.7	13.3	13.3	13.7	13.1	12.0	11.3	11.3	11.0
		MCWBR	7.6	7.5	7.7	7.0	6.3	6.3	7.1	7.1	6.0	6.2	7.0	7.0
	5% WB	MCDBR	9.8	10.5	11.1	10.2	11.1	11.9	11.7	11.7	10.4	9.6	9.7	9.1
		MCWBR	7.1	7.2	7.4	6.6	6.2	6.7	7.0	7.2	6.1	6.1	6.9	6.6
Clear-Sky Solar Irradiance	taub	0.352	0.376	0.424	0.461	0.463	0.447	0.447	0.455	0.454	0.427	0.388	0.358	
	taud	2.414	2.325	2.180	2.100	2.142	2.227	2.246	2.222	2.229	2.269	2.343	2.408	
	Ebn at Noon	824	846	834	823	827	838	835	819	796	778	769	787	
	Edh at Noon	88	108	137	157	153	141	137	137	129	113	94	83	
All-Sky Solar Radiation	RadAvg	2.29	3.06	4.32	5.49	6.56	7.35	7.57	6.65	4.94	3.58	2.51	2.03	
	RadStd	0.19	0.28	0.33	0.38	0.36	0.34	0.24	0.37	0.27	0.27	0.19	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	-1.62	-1.87	N/A	N/A	N/A	N/A	
Regional (9 neighbors)	N/A	N/A	N/A	N/A	-0.43	-0.61	N/A	N/A	N/A	N/A	

Nomenclature: See separate page